

A CONNECTED QUANDLE OF ORDER 70 ALL OF WHOSE MAXIMAL R -CLIQUES HAVE SIZE FOUR

ABSTRACT. For a rack $(Q, *)$ write $R_x(y) = y * x$, and call a subset $C \subseteq Q$ an R -clique if $[R_a, R_b] = 1$ for all $a, b \in C$. Kauffman, Mukherjee and Vojtěchovský ask whether there is a finite *connected* rack Q with a maximal R -clique C such that $|C|$ does not divide $|Q|$. We answer this affirmatively. Let Q be the set of the 70 three-cycles of A_7 with $x * y = y^{-1}xy$. Then Q is a connected quandle of order 70, every maximal R -clique of Q has size 4 — for instance $C = \{(1\ 2\ 3), (1\ 3\ 2), (4\ 5\ 6), (4\ 6\ 5)\}$ — and $4 \nmid 70$. The same argument gives, for every $m \geq 1$, a connected quandle of order $2m(36m^2 - 1)$ all of whose maximal R -cliques have size $4m$. We do not claim that 70 is the least order of such a quandle.

1. THE PROBLEM

Following [2], a *rack* is a set Q with a binary operation $*$ such that every right translation

$$R_y: Q \rightarrow Q, \quad R_y(x) = x * y,$$

is a bijection and such that $(x * y) * z = (x * z) * (y * z)$ for all $x, y, z \in Q$; a *quandle* is a rack that also satisfies $x * x = x$. The *right multiplication group* is $\text{Mlt}_r(Q) = \langle R_y : y \in Q \rangle \leq \text{Sym}(Q)$, and Q is *connected* if $\text{Mlt}_r(Q)$ acts transitively on Q . Write $a \sim b$ when $[R_a, R_b] = 1$, i.e. when $R_a R_b = R_b R_a$ as maps on Q . A subset $C \subseteq Q$ is an *R -clique* if $a \sim b$ for all $a, b \in C$, and it is *maximal* if no R -clique properly contains it. Section 5 of [2] studies quandles with many pairwise commuting right translations and closes with a list of problems. The following is one of them, quoted verbatim from the e-print arXiv:2504.09368v1 (line 1155 of the source file `main.tex`).

Problem (Kauffman, Mukherjee and Vojtěchovský). Does there exist a finite connected rack Q and a maximal R -clique C of Q such that $|C|$ does not divide $|Q|$?

The hypotheses are exactly *finite*, *connected* and *rack*. A quandle witness is a fortiori a rack witness, so we may and do produce a quandle.

2. THE WITNESS

For a finite group G and a subset $X \subseteq G$ closed under conjugation by X , the *conjugation quandle* on X is $(X, *)$ with $x * y = y^{-1}xy$; the construction is classical [1, 3]. We use one conjugacy class.

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Theorem 1. *Let Q be the set of all 70 three-cycles of A_7 , with $x*y = y^{-1}xy$. Then Q is a connected quandle of order 70, every maximal R -clique of Q has size 4, and in particular*

$$C = \{(1\ 2\ 3), (1\ 3\ 2), (4\ 5\ 6), (4\ 6\ 5)\}$$

is a maximal R -clique with $|C| = 4 \nmid 70 = |Q|$. Hence the question above has answer yes.

We prove the theorem in the following slightly more general form, which costs nothing extra.

Proposition 2. *Let $n \geq 5$ and let Q_n be the set of all three-cycles of A_n with $x*y = y^{-1}xy$. Then:*

- (a) Q_n is a quandle, and $|Q_n| = 2\binom{n}{3}$;
- (b) Q_n is connected;
- (c) for $a, b \in Q_n$ one has $a \sim b$ if and only if $ab = ba$ in S_n ;
- (d) distinct $a, b \in Q_n$ satisfy $ab = ba$ if and only if $\text{supp}(a) = \text{supp}(b)$, in which case $b = a^{-1}$, or $\text{supp}(a) \cap \text{supp}(b) = \emptyset$;
- (e) every maximal R -clique of Q_n has size $2\lfloor n/3 \rfloor$, and the maximal R -cliques are exactly the sets $\{a_1, a_1^{-1}, \dots, a_k, a_k^{-1}\}$ with $k = \lfloor n/3 \rfloor$ and $\text{supp}(a_1), \dots, \text{supp}(a_k)$ pairwise disjoint.

Proof. (a) The set of three-cycles of A_n is a union of S_n -conjugacy classes, hence closed under conjugation, so $*$ is defined. Each R_y is a bijection with inverse $R_{y^{-1}}$; $x*x = x^{-1}xx = x$; and self-distributivity is an identity of groups,

$$(x*z)*(y*z) = (z^{-1}yz)^{-1}(z^{-1}xz)(z^{-1}yz) = z^{-1}y^{-1}xyz = (x*y)*z.$$

A three-cycle is determined by its support together with one of the two cyclic orders on it, so $|Q_n| = 2\binom{n}{3}$; for $n = 7$ this is $2 \cdot 35 = 70$.

(b) The S_n -class of a three-cycle a does not split in A_n : since $n \geq 5$, at least two points lie outside $\text{supp}(a)$, and a transposition of two of them is an odd permutation centralising a . Hence Q_n is a single A_n -class. Now $\text{Mlt}_r(Q_n)$ is the image of the conjugation action of $\langle Q_n \rangle = A_n$ on Q_n , so its orbits are the A_n -classes contained in Q_n , of which there is one. Thus $\text{Mlt}_r(Q_n)$ is transitive.

(c) The composite $R_b R_a$ sends x to $a^{-1}b^{-1}xba = (ba)^{-1}x(ba)$, and $R_a R_b$ sends x to $(ab)^{-1}x(ab)$. These maps agree on Q_n if and only if $g := (ab)(ba)^{-1}$ centralises every element of Q_n , hence centralises $\langle Q_n \rangle = A_n$; as $Z(A_n) = 1$ for $n \geq 4$ this says $g = 1$, i.e. $ab = ba$. Conversely $ab = ba$ clearly gives $R_a R_b = R_b R_a$.

(d) Suppose $ab = ba$ with $a \neq b$. Then b maps $\text{supp}(a)$ onto itself, and b maps $\text{supp}(b)$ onto itself, so $\text{supp}(a) \cap \text{supp}(b)$ is b -invariant. As b is a three-cycle it acts transitively on $\text{supp}(b)$, so a b -invariant subset of $\text{supp}(b)$ is $\emptyset$ or $\text{supp}(b)$. In the first case the supports are disjoint. In the second, $\text{supp}(b) \subseteq \text{supp}(a)$ and both have three elements, so $\text{supp}(a) = \text{supp}(b) =: T$; then b centralises a inside $\text{Sym}(T) \cong S_3$, whose centraliser of the three-cycle

a is $\langle a \rangle = \{1, a, a^{-1}\}$, and $b \notin \{1, a\}$ forces $b = a^{-1}$. Both stated conditions do give commuting elements: permutations with disjoint supports commute, and a commutes with a^{-1} .

(e) Let S be an R -clique. By (c) and (d) the supports of the elements of S form a family of pairwise disjoint triples $T_1, \dots, T_k$, and S is contained in $\{a_1, a_1^{-1}, \dots, a_k, a_k^{-1}\}$ where $\text{supp}(a_i) = T_i$. Conversely any such set is an R -clique, again by (d). If S is maximal then $a_i^{-1} \in S$ for every i , since a_i^{-1} has support T_i and so commutes with every element of S ; hence $|S| = 2k$. Moreover at most two points of $\{1, \dots, n\}$ lie outside $T_1 \cup \dots \cup T_k$, for three uncovered points would carry a three-cycle disjoint from all of S , which could be adjoined. Therefore $k = \lfloor n/3 \rfloor$ and $|S| = 2\lfloor n/3 \rfloor$, and conversely every family of $\lfloor n/3 \rfloor$ pairwise disjoint triples yields a maximal R -clique. $\square$

Proof of Theorem 1. Apply Proposition 2 with $n = 7$: $Q = Q_7$ is a connected quandle of order $2\binom{7}{3} = 70$ and every maximal R -clique has size $2\lfloor 7/3 \rfloor = 4$. Since $70 = 4 \cdot 17 + 2$, no maximal R -clique divides the order. The displayed C consists of the two three-cycles on $\{1, 2, 3\}$ together with the two on $\{4, 5, 6\}$; these two triples are disjoint and $\lfloor 7/3 \rfloor = 2$, so C is one of the maximal R -cliques described in Proposition 2(e). $\square$

Corollary 3. *For $m \geq 1$ put $n = 6m + 1$. Then Q_n is a connected quandle of order $2\binom{n}{3} = 2m(36m^2 - 1)$, every maximal R -clique of Q_n has size $2\lfloor n/3 \rfloor = 4m$, and*

$$\frac{|Q_n|}{4m} = \frac{36m^2 - 1}{2} \notin \mathbb{Z}.$$

Proof. Immediate from Proposition 2, together with $\lfloor (6m + 1)/3 \rfloor = 2m$, the identity

$$2\binom{6m+1}{3} = \frac{(6m+1)6m(6m-1)}{3} = 2m(6m+1)(6m-1) = 2m(36m^2-1),$$

and the fact that $36m^2 - 1$ is odd. $\square$

3. VERIFICATION

The proof above is self-contained and uses no computation. The accompanying program `verify.py` independently re-derives, in exact arithmetic, the assertions of Theorem 1 about Q together with bounded cases of Proposition 2 and Corollary 3. It builds the 70×70 multiplication table of Q from the definition $x * y = y^{-1}xy$; confirms that each R_y is a permutation of Q , that $x * x = x$, and that there are 0 violations of self-distributivity over all $70^3 = 343000$ triples; confirms connectedness by computing an Mlt_r -orbit; checks that the table-level relation $R_a R_b = R_b R_a$ holds on exactly the pairs that commute in S_7 ; parses C from the cycle strings printed in Theorem 1 and certifies that it is a maximal R -clique; enumerates all maximal R -cliques of Q by Bron–Kerbosch, finding 70 of them, each of size 4; verifies Proposition 2(e) by direct enumeration of clique sizes and counts for $5 \leq n \leq 12$,

with the quandle axioms and the comparison against commuting in S_n re-run in full only for $5 \leq n \leq 8$; and evaluates the closed forms of Corollary 3 for $1 \leq m \leq 40$. The recorded run reports

VERDICT: ALL 85 CHECKS PASS

For $n \geq 13$ the program checks only closed-form arithmetic, and it enumerates no connected racks or quandles by order, so nothing in it bears on the least order of a witness.

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